

Period Tracker

A Command-Line Python Project

Project Report

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1. Introduction

Period Tracker is a simple command-line program written in beginner-level Python. It lets a user record the start date of each period, view their history, and get a prediction of when the next period is likely to start. The project was built to practise core Python skills while creating something useful, and it is published on GitHub as an open learning project.

2. Objectives

- Practise beginner Python concepts in a real project
- Store and reuse data between runs of the program
- Calculate an average cycle length and predict the next period
- Learn how to organise and publish a project on GitHub
- Protect user privacy by keeping all data on the user's own computer

3. Tools and Technologies

Tool	Purpose
Python 3	Main programming language
datetime module	Working with dates and calculating days between them
json module	Saving and loading data in a file
Git and GitHub	Version control and sharing the project
GitHub Actions	Automatically running tests on every push

4. Features

- **Log a period:** save a start date (press Enter to use today's date)
- **View history:** see every logged date in a numbered list
- **Predict next period:** uses the average of past cycle lengths
- **Delete last entry:** fix typos or mistakes
- **Input checking:** invalid dates are rejected with a friendly message
- **Persistent storage:** data is saved to periods.json

5. How the Program Works

The program runs a menu inside a **while** loop until the user chooses to quit. Each menu option calls its own function, which keeps the code easy to read.

Saving and loading data

Dates are converted to text (for example 2026-09-30) and stored in a JSON file. When the program starts, it reads the file and converts the text back into date objects. If the file does not exist yet, the program starts with an empty list.

Calculating cycle length

A cycle length is the number of days between one period start and the next. The program subtracts each date from the one after it:

```
def get_cycle_lengths(dates):
    lengths = []
    for i in range(1, len(dates)):
        lengths.append((dates[i] - dates[i - 1]).days)
    return lengths
```

Predicting the next period

The program averages all cycle lengths, rounds the result, and adds that many days to the most recent date. At least two logged periods are needed to make a prediction.

```
average = round(sum(lengths) / len(lengths))
next_date = dates[-1] + timedelta(days=average)
```

6. Project Structure

File	Description
period_tracker.py	The main program
test_period_tracker.py	Tests for the cycle-length calculation
periods.example.json	Fake sample data for demonstration
requirements.txt	States that no extra packages are needed
README.md	Project overview and instructions
DISCLAIMER.md	Medical disclaimer and privacy statement
CONTRIBUTING.md	How others can contribute
CHANGELOG.md	History of changes
LICENSE	MIT license
.gitignore	Keeps personal data and cache files off GitHub
.github/	Automated tests and issue templates

7. Sample Output

```
=== Period Tracker ===
```

```
1. Log a period
2. See history
3. Predict next period
4. Delete last entry
5. Quit
```

```
Choose an option: 3
```

```
Average cycle length: 28 days
```

```
Predicted next period: August 24, 2026
```

```
That was 37 day(s) ago - it may be late or need logging.
```

8. Testing

A small test file checks that cycle lengths are calculated correctly, including edge cases such as an empty list or only one date. The tests run with a single command (`python test_period_tracker.py`) and also run automatically on GitHub through GitHub Actions each time code is pushed.

9. Challenges and Learning

- Converting between date objects and text so they can be saved in JSON
- Handling wrong input with **try/except** so the program does not crash
- Splitting the program into small functions instead of one long script
- Making sure personal data is never uploaded to GitHub (using .gitignore)
- Learning GitHub basics: repositories, licences, and project documentation

10. Future Improvements

- Track symptoms and mood
- Record how many days each period lasts
- Show shortest and longest cycle
- Use only the last three cycles for a more accurate prediction
- Add a simple graphical interface or a mobile-friendly version

11. Disclaimer and Privacy

This project is a learning exercise and is **not** medical advice. Predictions are simple averages and may be inaccurate, especially for irregular cycles. It should not be used for birth control or health decisions. All data is stored only on the user's own computer, and the program never connects to the internet.

12. Conclusion

Period Tracker shows how a small idea can become a complete project using only beginner Python. It combines user input, loops, functions, file handling, and date calculations, and it was published with proper documentation, tests, and privacy considerations. The project provides a strong base for adding more features as skills improve.